(Please write your Enrollment Number) Enrollment No.

MID-TERM EXAMINATION
B. Tech. CSE-AI / ECE-AI/Reappear Semester: IV
(March, 2024) OFF LINE mode

Code: BAI 204	Optimization Techniques and Decision Making

Time : 1 ½Hours Maximum Marks : 30

Note: Q. 1 is compulsory.

Q1		(2.5*4)	
	(a) Explain Optimization, and its applications in Engineering		
	(b) Depending on whether a particular point belongs to the acceptable or unacceptable region, it can be identified as one of the four types. Define and explain these types?		
	(c) Explain the merits and limitations of the graphical method?		
	(d) Explain decision variables, objective function, and constraints. Write an LPP to illustrate these terms?		

Q2	(Attempt any Two Parts) UNIT-1	(5,5)
	(a) A soft drink manufacturing company has 300 ml and 150 ml canned cola as its products with profit margin of Rs. 4 and Rs. 2 per unit respectively. Both the products have to undergo process in three types of machine. The following data indicates the time required on each machine and the available machine-hours per week. Formulate the optimization problem as an LPP to maximize the total profit considering the limited resources.	

Requirement	Cola 300 ml	Cola 150 ml	Available machine hours per week
Machine 1	3	2	300
Machine 2	2	4	480
Machine 3	5	7	560

	(b) Discuss briefly about multiple and unbounded optimization Linear Programming Problems. Use appropriate example to justify your answer.	
	(C) Explain in detail the steps involved in formulating problems as mathematical programming problems? Explain the process, including the translation of design objectives and constraints into mathematical formulations	

Q3	(Attempt any Two Parts) UNIT-2	(5,5)
	(a) An airplane can carry a maximum of 200 passengers. A profit of Rs 1000 is made on each executive class ticket and a profit of Rs 600 is made on each economy class ticket. The airline reserves at least 20 seats for executive class. However, at least 4 times as many passengers prefer to travel by economy class than by the executive class. Determine using a graphical method how many tickets of each type must be sold in order to maximize the profit for the airline. What is the maximum profit?	
	(b) Use Simplex method to solve the following LP problem Maximize $Z = 50x + 60y$ subject to: $2x + y <= 300$; $3x + 4y <= 509$; $4x + 7y <= 812$; $x, y >= 0$.	
	(C) Write the algorithm to solve LPP using the simplex method OR explain the Integer Programming Problems in Optimization with an example.	